A Survey of Equatorial Waves

MEA 458

A quick comparison of weather variability - Tropics vs. Extratropics

- Have a look at the global forecast fields from tropical tidbits:

  Tropical Tidbits link
- What is the key difference in the variability of day-to-day weather in the tropics in comparison to the extratropics? Jot down your thoughts.

Tropics vs. Extratropics

- The key difference can be summed up in one sentence: In the tropics, the large scale fields of pressure, temperature etc. have very weak gradients in space and do not appear to change as rapidly as the extratropics.
- This does not mean that there is no weather variability in the tropics. There is. It occurs on synoptic to longer (intraseasonal) scales. This variability is best seen in fields of precipitating convection.

How does the animation of OLR look like? Some basic observations

Lets have a look at an animation of Outgoing Longwave Radiation (OLR) for a few days - this is a rather old link, but still works.

OLR animation

What are your observations based on the animations (jot them down. For now, focus on the top three panels. We will return to the fourth (sum of modes) in a short while)

How does the animation of OLR look like? Some basic observations

- OLR is a measure of deep, precipitating convection. In areas that are cloud free, the earth's long-wave radiation can escape to space. In these clear regions, OLR is high. In areas of deep clouds, the earth's radiation is absorbed by the clouds and the OLR is low. So, we can use OLR as a proxy for deep convection.
- The animation of OLR looks messy. We see clusters of convection that are moving eastward and westward. Some appear to be nearly stationary.
- When we remove the 40-day mean, the olr animation is cleaned up a bit - but it still looks messy. (Question: What does removing the mean OLR accomplish? What features are impacted?)

Making sense of tropical convective variability

- Is there a way to make sense of the hodge-podge of convection in the tropics?
- What if I told you that the seemingly messy convection in the tropics is actually a sum of distinct waves along with some stochastic convection.

Making sense of tropical convective variability

- Let me say that I processed the olr data and now I have these animations for you for the same days as seen in the previous animation. Have a look at the following link and write down your observations

  Wave modes

  You may wish to compare these animations to the ones seen earlier:

  OLR animation

Making sense of tropical convective variability

- Clearly, whatever processing we did of the original OLR, lo-and-behold, the convection now seems extremely tidy and well behaved.
- We now clearly see eastward and westward moving blobs of convection. The space between active convection is the suppressed areas
- one could draw an analogy with extratropical troughs and ridges - though active and suppressed convection may not necessarily coincide with troughs and ridges of waves in the tropics.

Making sense of tropical convective variability

So, What are these waves in the tropics?

We can broadly classify equatorial waves into:

- Rossby waves
- Kelvin waves
- Inertio-gravity waves
- Mixed Rossby-gravity waves

Equatorial Rossby waves

Figure 1: The theoretical equatorially-trapped Rossby wave solution. Hatching is for convergence and shading for divergence, with a 0.6 unit interval between successive hatching and shading, and with the zero divergence contour omitted. Unshaded contours are for geopotential, with a contour interval of 0.5 units. Negative contours are dashed and the zero contour is omitted. The dimensional scales are as in Matsuno (1966). (From Wheeler et al. 2000 via Roger Smith's notes).

Equatorial Rossby waves

A conceptual animation of Equatorial Rossby waves:

Link to animation from the MetEd textbook (Laing and Evans)

Equatorial Rossby waves: Observed

Figure 2: Evidence of equatorial Rossby waves in the Australian Bureau of Meteorology (BOM) 850hPa tropical wind analysis for 7 October 2002, 12 UTC. (Source: MetEd, Laing and Evans

Equatorial Rossby waves: Twin Tropical Cyclones

Figure 3: Double trouble: Tropical Cyclone Jamala (lower) and Tropical Cyclone 01B (upper storm) spin on opposite sides of the Equator in this infrared satellite image taken at 12 UTC (7 am EDT) Friday May 10, 2012. Image credit: University of Wisconsin CIMSS. Taken from: Weather Underground (link)

Mike Ventrice tweet on twin TCs

Figure 4: The theoretical equatorially-trapped Inertio-gravity waves. Hatching is for convergence and shading for divergence, with a 0.6 unit interval between successive hatching and shading, and with the zero divergence contour omitted. Unshaded contours are for geopotential, with a contour interval of 0.5 units. Negative contours are dashed and the zero contour is omitted. The dimensional scales are as in Matsuno (1966). (From Wheeler et al. 2000 via Roger Smith's notes).

Figure 5: The theoretical equatorially-trapped mixed Rossby-gravity waves. Hatching is for convergence and shading for divergence, with a 0.6 unit interval between successive hatching and shading, and with the zero divergence contour omitted. Unshaded contours are for geopotential, with a contour interval of 0.5 units. Negative contours are dashed and the zero contour is omitted. The dimensional scales are as in Matsuno (1966). (From Wheeler et al. 2000 via Roger Smith's notes).

Mixed Rossby-gravity waves: Observed

Figure 6: The enhanced IR satellite image of an equatorial mixed Rossby-gravity wave at 0000 UTC 21 November 2002 illustrates the impact of the convection on the wave structure. (Source: MetEd, Laing and Evans

Equatorial Kelvin wave

Figure 7: The theoretical equatorially-trapped mixed Kelvin wave. Hatching is for convergence and shading for divergence, with a 0.6 unit interval between successive hatching and shading, and with the zero divergence contour omitted. Unshaded contours are for geopotential, with a contour interval of 0.5 units. Negative contours are dashed and the zero contour is omitted. The dimensional scales are as in Matsuno (1966). (From Wheeler et al. 2000 via Roger Smith's notes).

Equatorial Kelvin waves

A conceptual animation of Equatorial Kelvin waves:

Link to animation from the MetEd textbook (Laing and Evans)

Equatorial waves

Why do we need to understand Equatorial waves?

- Tropical cyclones: Paper by Schreck, Molinari, Aiyyer
- Important part of tropical and extratropical weather forcing. Need to get the tropics right for extratropical weather prediction
- Review article by Kiladis et al.

Take a few minutes and email me a short summary of what you learned in this class and what was muddy.